

Emergency Evacuation Path Planner

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Introduction

The Emergency Evacuation Path Planner is designed to provide safe and efficient evacuation during emergencies. It helps people quickly find the nearest and safest exit routes in situations such as fires, earthquakes, or floods. The system uses intelligent path-planning algorithms to guide users effectively. Real-time monitoring helps detect hazards and blocked pathways. This project improves public safety and reduces confusion during emergencies. It is mainly useful in crowded places like malls, hospitals, airports, and office buildings.

Abstract :

- Developed to provide safe and shortest evacuation routes during emergencies.
- Uses path-finding algorithms such as Dijkstra's Algorithm and A* Algorithm.
- Detects hazards like fire, blocked paths, and congestion in real time.
- Dynamically updates evacuation routes based on current conditions.
- Helps reduce evacuation time and improve public safety.
- Useful in malls, hospitals, schools, airports, and office buildings.
- Assists emergency responders in managing evacuation effectively.
- Combines intelligent algorithms with real-time monitoring for disaster management.
- Contributes to the development of smart and secure infrastructure systems.

PEAS framework for AI evacuation planning

PEAS FRAMEWORK FOR AN AI AGENT (Emergency Evacuation Path Planner)

PEAS Element	Description	Emergency Evacuation Example	Implementation / Sensors
 Performance Measure	How well the agent achieves its goal.	- Minimize evacuation time - Maximize safety - Avoid hazards & congestion - Reach nearest exit	- Time taken to reach exit - Number of safe users evacuated - Hazard avoidance rate - Path optimality score
 Environment	The world in which the agent operates.	- Building layout (rooms, corridors) - Exits and emergency doors - Obstacles (fire, smoke, debris) - People / crowd movement - Dynamic hazards	- Building maps (BIM) - Fire & smoke sensors - Crowd density sensors - CCTV cameras - IoT environment sensors
 Actuators	The actions the agent can take.	- Suggest safest evacuation path - Update route in real-time - Send alerts & instructions - Guide to nearest exit	- Digital display / Mobile app - Audio alarms / Voice instructions - LED indicators / Signage - Notification system
 Sensors	The inputs the agent uses to perceive the environment.	- Detect fire, smoke, gas - Detect crowd density - Detect obstacles / blocked paths - Detect user location	- Smoke & gas sensors - Temperature sensors - Cameras / LiDAR - Wi-Fi / Bluetooth / GPS

Types of environments in AI system

(Emergency Evacuation Path Planner)			
Environment Type	Description / Observable	Agents	Example in Evacuation System
 Fully Observable	The agent has complete information about the environment.	Single / Multi	All exits, obstacles, people locations and hazards are fully known through sensors and maps.
 Partially Observable	The agent has incomplete information due to limitations in sensors or environment.	Single / Multi	Smoke or blocked paths may not be visible until the agent or sensors detect them.
 Deterministic	Environment is predictable; same action gives same result.	Single	If a path is safe now, it will remain safe if no changes occur.
 Stochastic	Environment is uncertain; actions may have probabilistic results.	Single / Multi	Fire spread, crowd movement or sensor noise can cause uncertainty.
 Static	Environment does not change while the agent is deliberating.	Single	Building layout and exits remain same during planning.
 Dynamic	Environment changes while the agent is deliberating or acting.	Single / Multi	People move, fire spreads, new obstacles may appear.
 Discrete	Environment changes at discrete time steps.	Single / Multi	System updates every few seconds to re-evaluate routes.
 Continuous	Environment changes continuously.	Single / Multi	Crowd density, smoke level or temperature change

Features

- Real-time path updates
- Hazard detection
- Alternative route suggestion
- Fast evacuation guidance

Applications

- Shopping malls
- Hospitals
- Schools and colleges
- Airports and railway stations
- Office buildings

Conclusion

The Emergency Evacuation Path Planner is an intelligent system developed to provide safe and efficient evacuation during emergency situations. It uses advanced path-planning algorithms and real-time monitoring to identify the shortest and safest routes for people. The system helps reduce evacuation time, avoid congestion, and improve public safety. It is highly useful in crowded places such as malls, hospitals, schools, airports, and office buildings. Overall, the project supports effective disaster management and contributes to the development of smart and secure infrastructure systems.

The background is a dark blue gradient. It features several abstract geometric patterns: a network of light blue lines with circular nodes in the corners, a horizontal line of small dots in the center, and a cluster of interconnected hexagons on the right side.

Thank You